Reinforcement Learning for Smart Grid Energy Routing to Mitigate Power Shortfalls during El Niño Events

ABSTRACT

Electricity scarcity creates a sequential allocation problem in which critical-service protection, storage availability, and the distribution of shortages across cities can conflict. This study formulates a daily energy-allocation controller for a synthetic Maharashtra case study comprising 15 cities and seven consumer categories. A proximal policy optimization agent selects three controls: battery charging or discharging, priority sharpness, and the strength of shortage rotation. A demand-capped weighted allocator maps these controls to 105 load allocations. The formulation separates aggregate energy feasibility from the learned policy and makes the resulting allocation preferences explicit. Analytical examination shows that changing priorities cannot reduce total unmet energy when supply is fixed and fully allocated, while finite priority weights do not guarantee complete service to critical loads. A reproducible numerical audit of the supplied allocator found no feasibility violations in 10,000 constructed cases at a tolerance of 10^{-7} MWh. An archived policy-comparison figure suggests lower aggregate and priority-weighted shortages for PPO than the displayed rules, alongside a fairness trade-off; exact effect sizes and statistical significance remain unestablished. The study provides a formal account of the implemented controller, preliminary evidence, and an evaluation protocol addressing forecast information, temporal continuity, storage accounting, and training variability. Its conclusions concern synthetic resource allocation rather than validated operation of an electrical network.

Keywords: reinforcement learning; proximal policy optimization; energy allocation; critical loads; battery storage; shortage fairness; El Niño; synthetic benchmark

1. INTRODUCTION

During an electricity shortfall, an allocation controller must decide which demands to serve and whether stored energy should be preserved for future periods. Serving all customers proportionally shares the shortage burden but gives the same relative curtailment to hospitals and less critical uses. Strong priority rules protect high-priority categories while potentially concentrating repeated shortages in particular cities. Storage introduces a temporal decision: discharging today may reduce the energy available during a more severe shortage tomorrow. These interactions motivate a controller that adapts allocation preferences and battery use together.

Climate variability provides a relevant scenario context. Ng, Turner, and Galelli [1] show that ENSO can influence regional hydropower production, with effects that vary geographically and can cancel at global scale. That evidence motivates testing energy controllers under climate-related stress; it does not establish a causal relationship between El Niño and every electricity deficit in Maharashtra. Here, ENSO labels, temperature, demand, and available energy are attributes of a synthetic dataset. Dates labeled 2026 identify a simulated evaluation period and must not be interpreted as measurements of an actual 2026 event.

The research question is whether a compact learned controller can improve priority-weighted service relative to credible storage-enabled rules while exposing the associated fairness costs. The project uses proximal policy optimization (PPO) [2] and a deterministic allocation layer. Its contribution is an application formulation and an inspectable control architecture, rather than a new reinforcement-learning algorithm. The technical analysis below identifies what this architecture guarantees, what it only encourages, and which experimental controls are necessary to assess the value of learning.

Three contributions are developed. First, the implemented allocation and battery transitions are expressed mathematically, including the distinction between city-history fairness and episode-level fairness. Second, analytical properties and a reproducible allocator audit establish the scope of aggregate feasibility and critical-load protection. Third, the study interprets the existing comparison conservatively and specifies a stronger evaluation design. These contributions support further empirical work without presenting unverified improvements as established results.

2. RELATED WORK AND RESEARCH POSITION

Reinforcement learning has been investigated for power-system decisions across control levels and operating states. Glavic, Fonteneau, and Ernst [3] review this breadth,

demonstrating that applying RL to an energy problem alone is insufficient to establish novelty. A useful contribution must identify the decision structure, represent the relevant operational constraints, and compare learning with strong alternatives. For this project, the distinctive design choice is a three-dimensional action that controls allocation preferences over many loads, rather than independently predicting all load allocations.

PPO uses a clipped policy-update objective to limit overly large changes between successive policies [2]. Stable-Baselines3 provides a common implementation framework [4]. These tools make the learning procedure practical, but the software library does not validate the physical assumptions or information available to the controller. In particular, hard energy limits enforced by a downstream allocator should be distinguished from penalties in a learned reward. The broader safe-RL literature explicitly distinguishes performance objectives from constraints that must be respected [5].

CityLearn v2 provides a useful reference for energy-control benchmarking because it supports comparisons among rule-based, model-predictive, and reinforcement-learning controllers and includes storage and resilience-related tasks [6]. Its building and community setting differs from the statewide synthetic allocation problem studied here. Nevertheless, its emphasis on a common environment motivates using identical storage, forecast, and accounting assumptions for all policies in the present evaluation. A change in the resource budget must not be credited to a learning algorithm.

The reliability of empirical RL comparisons is a further concern. Henderson et al. [7] document sensitivity to experimental choices and stochastic variability. Agarwal et al. [8] advocate uncertainty-aware evaluation rather than conclusions from point estimates alone. Accordingly, an archived bar chart is treated here as preliminary evidence. It does not establish consistency across independent training seeds, sensitivity to the synthetic-data generator, or improvement under information available at deployment.

Research position and limits of the present contribution : Table 1

Area	Relevant precedent	Present scope
Power-system RL	Decision and control across multiple operating states [3]	Daily aggregate allocation; no network power flow
Policy optimization	PPO and established implementations [2,4]	Existing PPO algorithm with three interpretable controls
Energy benchmarks	Common environments and multiple controller families [6]	Synthetic city and category allocation with storage
Reliability of evidence	Variability and interval estimation [7,8]	Analytical checks plus preliminary archived comparison

3. SYSTEM MODEL AND ALLOCATION PROBLEM

3.1 Loads and available energy

Let c index cities, k index consumer categories, i=(c,k) index loads, and t index daily transitions. The intended configuration has C=15 cities, K=7 categories, and N=105 loads. Let $d_{i,t} \geq 0$ denote demand in MWh and $p_{i,t} > 0$ denote a dimensionless priority weight. The seven categories are agriculture, commercial, educational institutions, hospitals, industry, public water works, and residential. The deployed scenario configuration assigns the weights in *Table 2*. These are modeling preferences; they are not established here as statutory government priorities.

Category priorities in the deployed scenario configuration : Table 2

Category	Priority	Critical in model
Hospitals	10	Yes
Public water works	9	Yes
Educational institutions	6	No
Residential	5	No
Commercial	3	No
Industry	2	No
Agriculture	1	No

The training environment reads priorities from the dataset. It defines critical loads using the first retained day and $p \geq 9$, so stable category priorities and consistent city-category ordering are required. For each day, the environment derives a state supply ratio ρ_t from statewide available energy Gt and statewide demand D_t . The modeled cities receive the same ratio of supply to demand:

$$\rho_t = \frac{Gt}{Dt} \ , \ S_t = \rho_t \, \Sigma i \, d_{i,t}. \quad (1)$$

This construction assumes a positive statewide demand and uniform exposure of the modeled cities to statewide scarcity. It does not model transmission capacity,

generation dispatch, electrical losses, voltage, frequency, or line outages. Consequently, "energy allocation" is the appropriate description; "routing" would imply network decisions that are absent. Allocations $x_{i,t}$ must satisfy $0 \le x_{i,t} \le d_{i,t}$ and $\Sigma_i\, x_{i,t} \le S_t + y_t$, where y_t is battery energy delivered to loads.

3.2 Battery dynamics

The battery is an aggregate model assumption with capacity B=50,000 MWh, a per-step energy-transfer limit R=10,000 MWh, and charge and discharge efficiency η=0.95. Because one transition represents one day, R is an energy-per-transition bound, not a specified instantaneous MW rating. The model initializes stored energy to $b_0 = \frac{B}{2}$. The implied charge-discharge round-trip efficiency is 0.9025. No degradation cost, self-discharge, or within-day operating constraint is represented. The first component of action a_t in $[-1,1]^3$ controls storage. If $a_{0,t} < -0.05$, delivered energy and the intermediate storage level are:

$$y_t = \min(-a_{0,t}R,\ \eta b_t),\quad b'_t = b_t - \frac{yt}{\eta}. \quad (2)$$

Otherwise $y_t = 0$ and $b'_t = b_t$. Allocation occurs before charging. With residual energy $v_t = \max(0, S_t + y_t - \Sigma_i x_{i,t})$, a positive command above 0.05 charges from residual supply only:

$$u_t = \min\left(a_{0,t}R,\ v_t,\ \frac{B - b'_t}{\eta}\right),\quad b_{t+1} = b'_t + \eta u_t . \quad (3)$$

For all other commands, $u_t = 0$. Charging and discharging cannot occur simultaneously. The controller cannot deliberately curtail currently served loads to charge the battery. These restrictions narrow the temporal scheduling problem and should be preserved in baseline comparisons.

3.3 Priority and shortage rotation

The remaining action components determine priority sharpness $\alpha_t = 1.5(a_{1,t}+1)$ in [0,3] and rotation strength $\beta_t = 0.5(a2,t+1)$ in [0,1]. Let $e_{c,t}$ be an exponentially smoothed city shortage percentage before the current allocation. Its standardized value $z_{c,t}$ is clipped to [−2,2] for the rotation factor, with z=0 when the cross-city standard deviation is negligible. Define:

$r_{i,t} = 1$ for critical loads; otherwise $r_{i,t} = \exp(0.5\beta_t\, z_{c(i),t})$. (4)

$w_{i,t} = d_{i,t}\, p_{i,t}^{\alpha t}\, r_{i,t}$. (5)

Recently underserved cities receive a larger rotation multiplier for noncritical loads. Critical loads have no direct multiplier, although their allocations can still change through competition for a shared budget. This distinction prevents the claim that removing critical loads from rotation makes their actual service independent of rotation. The water-filling routine begins with zero allocation and repeatedly distributes residual supply among loads with unmet demand in proportion to w. Each increment is capped at the remaining demand. It performs at most 12 passes and stops early if residual supply or the sum of eligible weights is negligible. With α=0 and β=0, weights are proportional to demand. With α=3, priorities are strongly emphasized but do not become lexicographic requirements.

3.4 Objective and fairness definitions

Let $\ell_{i,t} = d_{i,t} - x_{i,t}$. The percentage of weighted unmet demand is $W_t = 100\Sigma_i p_{i,t}\ell_{i,t}/\Sigma_i p_{i,t}d_{i,t}$. Let H_t be the percentage of unmet demand among critical loads. The city shortage percentage is $q_{c,t} = 100\Sigma_k \ell_{c,k,t}/\Sigma_k d_{c,k,t}$, assuming positive city demand. The history update and reward are:

$$e_{c,t+1} = 0.8e_{c,t} + 0.2q_{c,t},\ F_t = \mathrm{SD}_c(e_{c,t+1}). \quad (6)$$

$$R_t = -(W_t + 2H_t + 0.5F_t). \quad (7)$$

These coefficients express a chosen trade-off and are not derived from economic valuation. Reward fairness uses the dispersion of smoothed shortage percentages. Evaluation instead reports SDc ($100\Sigma_{t,k}\ell_{c,k,t}/\Sigma_{t,k}d_{c,k,t}$), the dispersion of episode-wide city shortage fractions. Both are measured in percentage points, but they answer different questions. A policy can perform well on one and poorly on the other. The unnormalized priority-weighted shortage is a weighted-energy score with MWh multiplied by dimensionless priority, not additional physical energy.

4. LEARNING ARCHITECTURE AND ANALYTICAL PROPERTIES

4.1 Observation and learning procedure

The observation has 26 components: seven global features, 15 city histories, and four previous shortage fractions for

the highest priority tiers. Global features include the current supply ratio minus one; mean and minimum forecast supply ratios minus one; battery state of charge; normalized Oceanic Niño Index; normalized temperature; and relative statewide demand. City histories are divided by ten. The complete observation is clipped to [−10,10], then normalized by VecNormalize for policy inference. The forecast features are constructed from the next seven retained supply-ratio entries with independent Gaussian perturbations of standard deviation 0.01. They are future-informed synthetic features, not outputs of a forecast model fitted using past data. Near the sequence end, fewer entries are used, with the current ratio as fallback. Filtering to El Niño before forming this window means seven entries need not represent seven consecutive calendar days. Relative statewide demand is normalized using the mean of the selected split, which also introduces information from later observations within validation or test.

The full simulator state includes the date index, exogenous trajectories, storage, and shortage histories. The compressed observation does not explicitly contain the complete current city-category demand vector. Consequently, a feed-forward policy receives a partially observed control problem, and the 26-component vector should not be asserted to be a sufficient Markov state. This can limit adaptation when the composition of demand changes without a corresponding change in aggregate features.

PPO [2] optimizes a clipped surrogate objective using policy probability ratio $r_t(\theta)=\pi_\theta\ (a_t\ |o_t\ )/\pi_{old}(a_t\ |o_t\ )$ and estimated advantage $\hat{A}_t$:

$$L^{CLIP}(\theta) = E[\min(r_t(\theta)\hat{A}_t\ , \text{clip}(r_t(\theta),1-\varepsilon,1+\varepsilon)\hat{A}_t\ )]. \quad (8)$$

This is the policy component of the PPO training objective; the implementation also fits a value function. The project uses the Stable-Baselines3 MlpPolicy, observation and reward normalization during training, and deterministic actions during evaluation. *Table 3* records explicit script settings. The 500,000-step value is a requested training budget; exact completed transitions require the training log because rollout-based implementations may finish a complete rollout beyond the requested count.

Explicit training configuration : Table 3

Setting	Configured value
Training seed	42
Requested transitions	500,000
Parallel environments	4
Training window	60 retained daily entries; contiguous starts preferred
Rollout steps per environment	1,024
Minibatch size and epochs	256 and 10
Learning rate	0.0003
Discount and GAE parameter	0.99 and 0.95
PPO clipping and entropy coefficient	0.2 and 0
Validation interval	10,000 aggregate environment transitions
Validation episodes per check	1 full validation sequence

4.2 Aggregate feasibility

Proposition 1. Assume finite nonnegative supply and demands, and nonnegative weights with positive total weight on eligible loads whenever allocation proceeds. Each water-filling pass preserves nonnegative allocation, the per-load demand caps, and the aggregate supply cap in exact arithmetic. To see this, every increment is nonnegative and no larger than the remaining demand. Before demand capping, the increments sum to the residual supply; applying the caps can only reduce that sum. Induction over the passes establishes the claim. Stopping after a finite number of passes preserves feasibility but is not a universal proof that all allocatable supply has been used. Floating-point computations require a numerical tolerance.

Proposition 2. If the allocator uses $\min(S_t\ +y_t\ ,\Sigma_i d_{i,t})$ energy, then total daily shortage equals $\max(0,\Sigma_i d_{i,t}-S_t\ -y_t\ )$. Therefore, for a fixed battery output, changing priority sharpness or rotation changes the distribution of shortage but cannot change its total. A reduction in total shortage relative to another policy must arise from storage scheduling, differences in wasted supply, or a changed model or accounting convention. This identity is a useful consistency check on reported policy comparisons.

Proposition 3. Finite priority weights do not guarantee zero shortage for critical loads even when available energy is sufficient to meet critical demand alone. Consider one critical and one noncritical load, each demanding 100 MWh, priorities 10 and 1, supply 100 MWh, no battery, and no rotation. At $\alpha=3$, the allocator gives the critical load $\frac{100\times1000}{1001}\approx99.9001$ MWh. A strict critical-first allocator

could give it the full 100 MWh. The present method provides strong preference, not a hard minimum-service guarantee.

5. EXPERIMENTAL EVIDENCE AND REPRODUCIBILITY

5.1 Data provenance and temporal split

The project identifies its data source as the Maharashtra Electricity Dataset (Synthetic), hosted on Kaggle [9]. The repository describes it as artificial data for academic research and identifies a CC BY 4.0 license. The data are not official electricity records. The environment expects date, city, category, demand, priority, statewide supply and demand, ONI, temperature, and ENSO-phase fields. Complete city-category grids, unique keys, finite values, positive denominators, and consistent daily metadata are necessary before the current reshape-based loading procedure can be trusted. The configured chronological split uses dates before 2024 for training, 2024-2025 for validation, and dates from 2026 onward for testing. The intended dataset spans 2015-2026; the test condition itself does not impose an upper bound. Training samples 60-day windows whose endpoints belong to the same run of consecutive dates. If no such window exists, the fallback starts at the beginning and may cross gaps. Validation and test traverse all retained El Niño dates in order, carrying storage and shortage histories across any gaps. This differs from elapsed calendar-time evolution. The numerical dataset, per-day evaluation trajectories, and the comparison CSV are required to reproduce the full policy experiment. The present empirical record consists of saved policy artifacts, the implementation, and an archived comparison image. The image does not establish the exact number of test days, the checkpoint used, or the statistical variability. These omissions limit the empirical conclusions to the qualitative ordering visible in that figure.

5.2 Baseline definitions

Rules configured in the offline evaluator : Table 4

Policy	α	β	Battery rule
Proportional allocation	0	0	Idle
Priority rule	3	0	Idle
Priority and greedy battery	3	0	Greedy
Priority battery and rotation	1.5	1	Greedy
PPO	Learned	Learned	Learned

The greedy rule discharges at the maximum action when the current supply ratio is below one and requests charging otherwise; physical limits and surplus-only charging still apply. The rotation baseline changes both α and β relative to the priority-plus-battery rule. Their difference therefore does not isolate rotation. A controlled rotation ablation must hold α fixed. The offline evaluator defaults to the final policy, while an optional path selects the best validation checkpoint. The normalization state saved at the end of training is not necessarily the correct normalization state for an earlier best checkpoint.

5.3 Reproducible allocator audit

An additional numerical audit directly executes the waterfill function extracted from the environment source without changing that function. It uses NumPy random seed 20261103 and constructs 10,000 independent 105-load cases. Each demand is sampled uniformly from 1 to 1,000 MWh, supply is sampled from zero to 1.5 times aggregate demand, and α is sampled uniformly from zero to three. The seven category priorities in Table 2 repeat for each city. Each city receives a log-rotation multiplier sampled uniformly on $[-1,1]$ and exponentiated; critical-load multipliers are set to one. This explores the multiplier envelope without claiming that every sampled vector is reachable from a single standardized city-history state.

At a tolerance of 10^{-7} MWh, the audit found zero negative allocations, demand-cap violations, or aggregate supply-cap violations. The largest unused allocatable supply was 1.46×10^{-11} MWh, consistent with floating-point residuals. These tests check the standalone allocator on a declared constructed distribution. They do not test PPO performance, battery trajectories, historical Maharashtra data, or robustness to malformed inputs. The finite-pass implementation may still need a residual-tolerance termination rule for other weight distributions.

Constructed two-load example using the supplied allocator : Table 5

Sharpness α	Critical served MWh	Other served MWh	Weighted shortage

0	50.0000	50.0000	550.0000
1	90.9091	9.0909	181.8182
1.5	96.9347	3.0653	127.5881
3	99.9001	0.0999	100.8991

In every row of *Table 5*, total shortage is exactly 100 MWh up to numerical precision. Raising α redistributes this shortage away from the critical load, decreasing the weighted shortage without creating energy. The example verifies the interpretation of Propositions 2 and 3. It is a deliberately constructed illustration, not a reconstruction of the project dataset.

6. PRELIMINARY POLICY COMPARISON

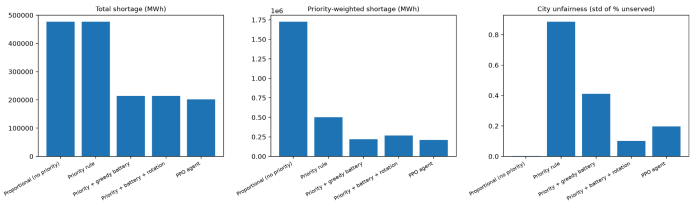


Figure 1. Archived project comparison associated with the 2026 synthetic evaluation. The original plot is reproduced without extracting exact numerical values. The middle axis denotes a priority-weighted energy score; the right axis denotes dispersion in percentage points. Run provenance and uncertainty intervals are not available in this figure.

Figure 1 shows the PPO bar below the four displayed rules for total shortage and priority-weighted shortage. The two rules without storage show approximately equal total shortage, while the storage-enabled rules show substantially lower total shortage. This pattern is consistent with the conservation identity in Proposition 2: prioritization can change who experiences a deficit, whereas storage can shift energy between days. The appropriate comparator for the incremental benefit of PPO is therefore a tuned controller with the same battery and information, not solely a rule without storage. The fairness panel gives a different ordering. Proportional allocation has approximately zero cross-city dispersion, and the battery-plus-rotation rule has lower dispersion than PPO. PPO appears fairer than the priority-only rule and the priority-plus-greedy-battery rule. This supports interpreting the controller as a compromise between weighted service and geographic fairness rather than as uniformly superior. Proportional allocation can be very equal while still imposing unacceptable critical-load shortages, so fairness alone is not an adequate welfare measure.

No exact percentage gain, confidence interval, critical-load performance estimate, or significance test is inferred from the image. The offline evaluator defines two additional metrics, critical-load shortage and average weighted unmet demand, that are not displayed. Their values cannot be recovered reliably from the three plotted measures. Similarly, the configured reward combines daily percentages and smoothed fairness, so a lower total weighted-shortage bar does not by itself prove a higher cumulative reward.

The strongest empirical statement supported at this stage is that the archived figure is consistent with a promising storage and allocation trade-off. It does not establish that PPO reliably exceeds the strongest baseline, that it generalizes to unseen climate regimes, or that the result would persist when future-informed features are replaced with causal forecasts. Exact results must be regenerated from the identified data version and policy-normalizer pair before a definitive performance claim is made.

7. EVALUATION PROTOCOL FOR A CONFIRMATORY STUDY

7.1 Information and resource parity

A confirmatory experiment should first freeze the dataset version, source revision, controller definitions, and primary metric. All methods should receive the same permissible information and identical initial storage, capacity, efficiency, rate limits, and end-of-horizon accounting. The primary comparison should use cumulative priority-weighted shortage, with critical-load shortage, total energy not served, fairness, and worst-city service reported separately. Report both fairness definitions rather than substituting one for the other.

Forecasts must be generated without access to future test observations. Compare persistence, a forecasting model fitted only on training data, and a clearly labeled future-informed reference. Fit feature scaling on training data and freeze it for validation and test. A useful upper reference is a perfect-information finite-horizon dispatch optimization under the same physical constraints. It should be labeled an oracle bound or reference, not a deployable baseline. A forecast-based model-predictive controller then tests whether learning adds value over explicit sequential optimization.

The battery should either end at its initial state of charge or receive a transparently defined terminal-energy accounting rule. Otherwise, a policy can improve the measured horizon by consuming energy made available free at reset. At 50% initial charge, the model starts with 25,000 MWh stored and can deliver up to 23,750 MWh of that energy before accounting for subsequent charging. A learned policy should not receive credit for a resource unavailable to its comparator.

7.2 Baselines and ablations

Retain the existing rules but tune their sharpness, rotation, and battery thresholds using validation only. Add a strict critical-first allocation rule, a tuned static allocation rule with greedy storage, and a forecast-based model-predictive controller. Compare a second continuous-control RL method only after the common environment and information protocol are validated. Use the same compute or tuning budget where practical, and disclose differences.

Ablate rotation while holding priority sharpness fixed, freeze battery control while keeping the learned allocation policy, and remove forecast features. Separately remove ONI and temperature to test whether explicit climate covariates add predictive control value beyond supply and demand. Retraining is needed for feature-removal and action-removal ablations; masking a feature only at inference measures distribution shift as well as information loss. Test different battery capacities, efficiencies, shortage severities, demand mixes, and reward coefficients. Present a service-fairness frontier rather than choosing a favorable coefficient after seeing test results.

7.3 Statistical reporting

Use at least ten independently trained seeds as a practical starting point, with paired evaluation scenarios and forecast-noise realizations across methods. This number is a proposed design choice, not a guarantee of statistical power. Determine whether more runs are necessary from the width of uncertainty intervals. Keep training-seed variation distinct from scenario variation; evaluation on many days from one trained model does not replace independent training runs.

For each comparator b, define the paired change $\Delta b = J_{PPO} - J_b$, where J is the primary lower-is-better metric. Report the distribution and a 95% interval for Δ, together with the raw metric values. If reporting a percentage improvement, define it as $100(J_b - J_{PPO})/J_b$ and avoid it when the denominator is zero. Preserve temporal dependence when resampling trajectories, for example using whole episodes or contiguous blocks. Scenario-level replication must reflect genuinely independent generated trajectories, not repeated samples from one deterministic path. This design follows the general emphasis on uncertainty in [7,8].

8. LIMITATIONS AND DEPLOYMENT IMPLICATIONS

Synthetic-data dependence is the primary external-validity limitation. A generator can encode a relationship between ENSO, temperature, and supply that an agent then exploits, without demonstrating that the relationship holds in observed operation. Publishing the generation procedure, calibration targets, and random seeds is necessary to make that dependence inspectable. Independent observed data would support a stronger applied claim; otherwise, the contribution should remain explicitly synthetic and methodological.

The daily aggregation omits intra-day peaks and cannot demonstrate voltage, frequency, or network-security performance. The chosen priority scale and critical-load definition express assumptions that require domain justification. Finite priority weights and a critical-shortage penalty do not establish a hard service guarantee. An operational extension would need explicit minimum-service constraints, an infeasibility policy when critical demand exceeds available supply, and evaluation of starvation in lower-priority categories.

Temporal filtering and normalization can materially affect results. Carrying storage across separated ENSO segments compresses elapsed time, and retrospective forecast features provide privileged future information. These issues should be corrected or clearly separated into experimental conditions before estimating practical performance. In addition, a best-checkpoint evaluation must restore observation statistics saved with that checkpoint, rather than assume the final normalization state is interchangeable.

The dashboard is a scenario demonstrator with a different information regime from the multi-day environment. It initializes storage and shortage histories for each request, substitutes fixed climate context, and does not demonstrate persistent rotation over a chronological rollout. Its response path also caps final allocation at the original external

supply after the environment may have added discharged battery energy. Some returned metrics are computed before that rescaling. This accounting difference must be reconciled before dashboard outputs are used as evidence for the offline method. The dashboard priority baseline uses a different priority exponent from the offline baseline, adding another comparability issue.

These limitations identify concrete research controls rather than reasons to discard the architecture. Separating a learned preference policy from an aggregate feasibility layer remains useful for inspection and experimentation. However, network feasibility, minimum critical service, causal climate information, and robust comparative performance are separate claims that require separate evidence.

9. CONCLUSION

This study formalizes a compact PPO controller for battery operation and priority-sensitive allocation in a synthetic 15-city electricity-scarcity setting. The demand-capped allocator enforces aggregate supply and demand bounds under stated input assumptions. Analytical results establish that allocation preferences redistribute shortage, while storage scheduling can change the total energy served, and that finite priority weights do not guarantee uninterrupted critical-load service. A 10,000-case constructed audit supports the numerical feasibility of the allocator within its tested input envelope.

The archived comparison suggests a favorable shortage trade-off for PPO but also shows that lower shortage and greater geographic equality are distinct outcomes. The evidence does not yet establish statistically reliable improvement or real-grid applicability. A credible confirmatory study requires causal forecasts, chronological evaluation, consistent battery accounting, strong optimization baselines, paired multi-seed experiments, and complete data and artifact provenance. With these controls, the architecture can serve as a transparent testbed for studying the value of adaptive priority and storage decisions under climate-conditioned synthetic scarcity.

Data and code availability

The dataset source is the Kaggle entry in [9], described by the project as synthetic and licensed under CC BY 4.0. The implementation comprises the SmartGridEnv environment, PPO training script, offline evaluator, and scenario API. The allocator audit accompanying this manuscript records its seed, numerical checks, constructed cases, and a SHA-256 fingerprint of the environment source. A public archived code release and a versioned data snapshot are needed for complete external replication. No observed Maharashtra grid records are analyzed in this study.

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